4 Reference Frames

Question 1

Consider a pure turbojet airplane flying under cruising conditions at an altitude of $h_G = 30000\,\text{ft}$ and a speed of $500\,\text{mph}$. The engine has an inlet of $A_i = 7.0\,\text{ft}^2$ and an outlet of $A_e = 4.5\,\text{ft}^2$. The speed and pressure of the exhaust gases at the outlet are $V_e = 1600\,\text{ft/s}$ and $p_e = 640\,\text{lb/ft}^2$, respectively. Compute the thrust of the turbojet and its efficiency.

Question 2

Consider the aircraft of **Question 1**.

- Compute the components of the velocity vector in the Aerodynamic Frame, in the Body Frame, and in the Vertical Frame, if the angle of attack is $\alpha = 5°$, the angle of sideslip is $\beta = 1°$ and the climb angle is $\gamma = 3°$.
- If the aircraft weighs $W = 10100\,\text{lb}$ and the angles of elevation and bank are $\Theta = 10°$ and $\Phi = 20°$, respectively, compute the components of the weight in the Body Frame and in the Wind Frame.

Question 3

Consider the data of **Question 1** in Assignment (3) and the conditions of gliding flight from $h_G = 10000\,\text{ft}$.

- Compute the maximum range and the maximum endurance.
- Compute the V_{KTAS} needed to achieve the maximum range and the maximum endurance and briefly discuss the difference between the obtained results.